

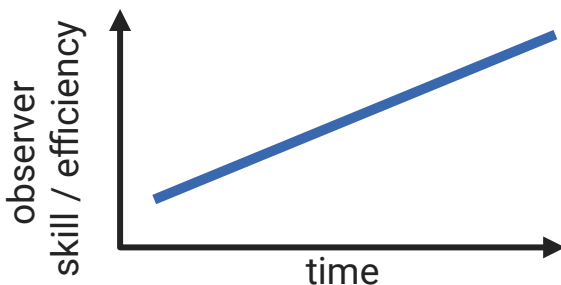
## GLM / GLMM

combined abundance/  
detection model:  
 $y_i \sim \text{Poisson}(\lambda_i)$

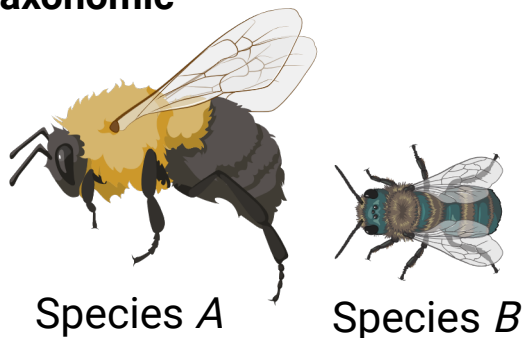
potential detection confounds:  
spatial



temporal



taxonomic



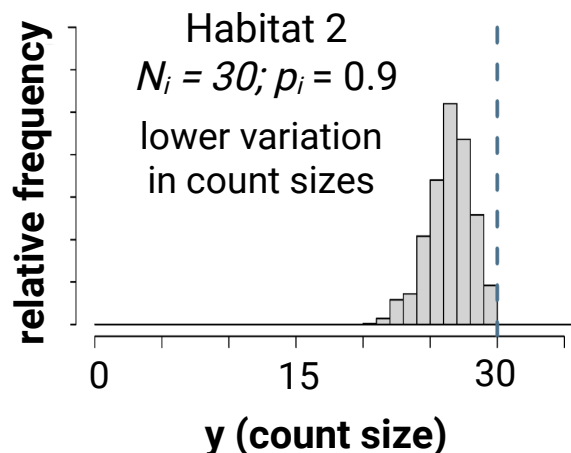
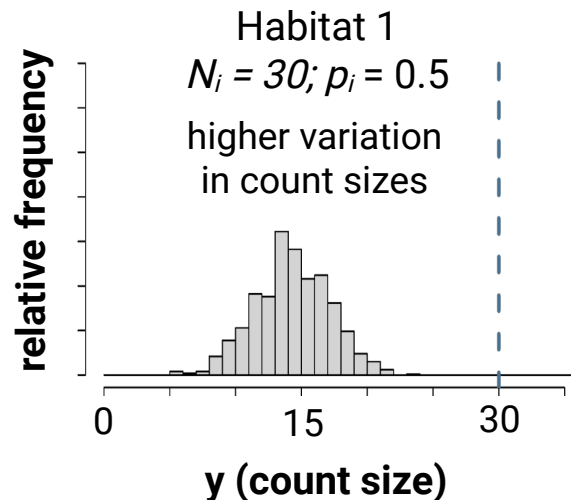
## Binomial N-mixture

abundance model:

$$N_i \sim \text{Poisson}(\lambda_i)$$

detection model:

$$y_{i,j} \sim \text{binomial}(N_i, p_i)$$



## Multinomial N-mixture

abundance model:

$$N_i \sim \text{Poisson}(\lambda_i)$$

detection model:

$$n_i \sim \text{binomial}(N_i, 1 - \pi_0);$$

$$y_i | n_i \sim \text{multinomial}(n_i, \pi_i^c)$$

